

GIRTH-12 LIFTS OF THE VOLTAGE GRAPH G_{12} FOR $m = 6, 7, 8$, AND NONEXISTENCE ON ITS PRINTED ARC SET FOR $m = 4, 5$

ABSTRACT. Aguilar, Araujo-Pardo and Berman construct a voltage graph G_{12} whose $\mathbb{Z}_m$ -lift is a $(\{3, m\}; 12)$ -graph of order $49m + 3$ for every $m \geq 9$, and ask, in the second sub-bullet of their open-questions section, for voltage assignments producing $(G_{12}; m)$ -graphs for $4 \leq m \leq 8$ or a proof that none exist. For $m = 6, 7, 8$ we exhibit voltage assignments on the arcs of G_{12} whose lifts are $(\{3, m\}; 12)$ -graphs of orders 297, 346, 395; that answers the sub-bullet at those three values of m . Separately, with the 24 arcs of their Table 3 held fixed as printed — with the evident correction to the duplicated starting leaf — and only the 24 voltages varying, we prove that no assignment gives girth 12 at $m = 4$ or at $m = 5$: at $m = 4$ by a four-variable pigeonhole argument requiring no computation, four specific arc voltages having to be pairwise distinct and nonzero while $\mathbb{Z}_4$ has only three nonzero residues; at $m = 5$ by exhaustion of the complete cell $\mathbb{Z}_5^{24}$. We do not settle whether the sub-bullet asks only about this fixed arc set or about every girth-12 graph obtainable from the same underlying tree; under the wider reading the cells $m = 4, 5$ remain open, so what is proved here at $m = 4, 5$ is a fixed-arc statement. No order record and no improved upper bound is claimed at any m : the published bounds dominate all three witnesses.

1. THE QUESTION

Write $n(\{r, m\}; g)$ for the least order of a graph all of whose degrees lie in $\{r, m\}$, in which both degrees occur, and whose girth is g ; such a graph is a $(\{r, m\}; g)$ -graph. Aguilar, Araujo-Pardo and Berman [1] study these for $r = 3$ (*semicubic*) using voltage graphs. Throughout their paper the voltage group is cyclic: “*Throughout this paper, Γ is a cyclic group $\mathbb{Z}_m$ with addition as the group operation*” [2].

Their Theorem 5.5 reads, verbatim and identically in the preprint [2] and in the version of record [1, p. 25]:

“The graph $(G_{12}; m)$ in $\mathcal{G}_{4t}$ for $t = 3$ given by the voltage graph G_{12} formed by adding the arcs and voltage assignments to T_{12} given in Table 3 has girth 12 for $m \geq 9$. Moreover, it is a $(\{3, m\}; 12)$ -graph with three vertices of degree m and $49m$ vertices of degree 3.”

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The question addressed here is the **second sub-bullet** under the bullet “Related to semicubic graphs” of their Section 6, “Open questions”, verbatim from [1, p. 29] and, identically, from the preprint [2] (the “ $\leq$ ” is the PDF’s glyph and the subscripts are flattened as in the source’s own inline style):

“Find voltage assignments to produce $(G_{12}; m)$ graphs with the construction given in Theorem 5.5 for any $4 \leq m \leq 8$, or show no such graphs exist.”

By Theorem 5.5 the object sought for a given m has order $49m + 3$, that is 199, 248, 297, 346, 395 for $m = 4, \dots, 8$, with three vertices of degree m and $49m$ of degree 3, and is the $\mathbb{Z}_m$ -lift of G_{12} under *some* voltage assignment on the 24 arcs of Table 3.

Theorem 1. *Let G_{12} be the graph of Section 2 — the tree T_{12} of Table 1 together with the 24 arcs of Table 2. For each $m \in \{6, 7, 8\}$ there is a $\mathbb{Z}_m$ voltage assignment on those 24 arcs whose lift is a $(\{3, m\}; 12)$ -graph of order $49m + 3$, namely 297, 346, 395; one such assignment is exhibited for each m in Section 4.*

Theorem 2. *Let the 24 arcs of Table 2 be held fixed as printed, so that only the 24 voltages vary. Then no $\mathbb{Z}_m$ voltage assignment on those arcs has a lift of girth 12 for $m = 4$ (Proposition 6, a four-line argument with no computation), nor for $m = 5$ (Section 6, by exhaustion of all 5^{24} assignments).*

Scope of Theorem 2. Theorem 1 needs no caveat: the assignments it exhibits live on the printed arcs, which the sub-bullet permits on any reading, so at $m = 6, 7, 8$ the sub-bullet is answered outright. Theorem 2 is narrower — the 24 arcs are held *fixed as printed* — and *we do not settle that this is the reading its authors intended*. What can be said for it is this: Theorem 5.5 defines G_{12} as T_{12} plus “the arcs **and voltage assignments** . . . given in Table 3”, so the arcs are part of the named object, and [1] poses a re-arc-ing question in a *separate* sub-bullet, about a different graph H'_{12} . That is an inference about wording and not a proof of intent; no author clarification was obtained. Accordingly Theorem 2 must always be read as “no $\mathbb{Z}_m$ voltage assignment on the arc set of G_{12} gives girth 12” at $m = 4, 5$, and never as the unqualified “no such graphs exist”. Under the wider reading, in which each x -leaf may be re-paired to some y -leaf and some z -leaf, the cells $m = 4, 5$ are untouched here and remain open; see Section 8. The group is $\mathbb{Z}_m$ and nothing else, by the sentence quoted above; in particular the $m = 4$ cell is $\mathbb{Z}_4^{24}$ and not also $(\mathbb{Z}_2 \times \mathbb{Z}_2)^{24}$.

Theorem 5.5 is not weakened by any of this. With the printed Table-3 voltages the lift has girth 6, 6, 8, 8, 8, 10 at $m = 3, \dots, 8$ and 12 at $m = 9, \dots, 14$ (recomputed by the program of Section 7), so its threshold $m \geq 9$ is exactly attained for the object it names — a voltage graph includes its voltages — and what Theorem 1 exhibits are *other* assignments on the same underlying graph.

No record is claimed. The published upper bounds at the three cells where we exhibit a graph are $n(\{3, 6\}; 12) \leq 243$, $n(\{3, 7\}; 12) \leq 334$ and $n(\{3, 8\}; 12) \leq 374$, as tabulated by Goedgebeur, Jookan and Van den Eede [3, girth-12 rows of its bound table]; the last is Corollary 3 of [1] itself. Our three witnesses have orders 297, 346, 395 and are dominated in every case. The source’s own introduction says of this family that “*the order of these graphs is bigger than the upper bounds from the identifying vertices construction*”; a reading of the witnesses as improvements would contradict it.

Adjacent work. [3] generalises the gluing theorem of [1], and its girth-12 table gives bounds for exactly the five cells at issue, the bound 243 at $m = 6$ among them; but it uses no voltage graphs or lifts, and it does not address the existence of a girth-12 $\mathbb{Z}_m$ voltage assignment on the arc set of G_{12} , which is the question here. Exoo and Jajcay [4] give girth criteria for $\mathbb{Z}_m$ lifts of voltage graphs, the closest work methodologically, but treat neither G_{12} nor these five cells; Exoo [5] searches voltage assignments for regular cages, a related technique on different objects. Araujo-Pardo, Ramos-Rivera and Jajcay [6] and Araujo-Pardo, Kiss and Szőnyi [7] reach girth-12 bipartite biregular cages through generalized hexagons and designs, again different objects, and settle nothing about G_{12} .

2. THE GRAPH G_{12} , WRITTEN OUT IN FULL

The source gives T_{12} by prose and the arcs by its Table 3. Both are transcribed here completely, so that no figure, no external file and no program is needed to read what follows. Throughout, “Table 3” with no further qualification means the source’s table; the tables of the present note are Tables 1 and 2, and Table 2 is the source’s Table 3 with the correction of Remark 2 applied.

T_{12} has 52 vertices — 49 ordinary ones and three *pinned* vertices x^*, y^*, z^* of degree 1 — and 51 edges, all carrying voltage 0. A pinned vertex is a degree-1 base vertex: in the lift a *single* vertex x^* is joined to all m copies of its unique neighbour, so $\deg(x^*) = m$ and the voltage on that edge is irrelevant.

Equivalently, and this is how [1] defines it: the vertices of X'_3 are the binary strings of length at most 4 with every string beginning 00 deleted except 00 itself (so x_{00} is a leaf of the tree part), plus the pinned x^* above the root; the vertices of $Y_2 = Z_2$ are the binary strings of length at most 3 with every string beginning 00 deleted, plus y^* ; and x_{00} is joined to y_0 and to z_0 . The twelve depth-5 leaves $x_{0100}, \dots, x_{1111}$ are exactly the twelve “Starting leaf” entries of Table 3, and the six leaves of each of Y_2, Z_2 are its “End” entries.

The symbol names $a, b, \dots, \varepsilon$ (the list skips i, m, n, o) are the authors’ own, taken from a commented-out substitution list in the preprint’s source. Reading the 24 symbols in the interleaved order above reproduces both

TABLE 1. The 51 tree edges of T_{12} , all of voltage 0.

X'_3 (26 vertices, 25 edges)	x^*x ; xx_0, xx_1 ; x_0x_{00}, x_0x_{01} ; x_1x_{10}, x_1x_{11} ; $x_{01}x_{010}, x_{01}x_{011}$; $x_{10}x_{100}, x_{10}x_{101}$; $x_{11}x_{110}, x_{11}x_{111}$; $x_{010}x_{0100}, x_{010}x_{0101}$; $x_{011}x_{0110}, x_{011}x_{0111}$; $x_{100}x_{1000}, x_{100}x_{1001}$; $x_{101}x_{1010}, x_{101}x_{1011}$; $x_{110}x_{1100}, x_{110}x_{1101}$; $x_{111}x_{1110}, x_{111}x_{1111}$
Y_2 (13 vertices, 12 edges)	y^*y ; yy_0, yy_1 ; y_0y_{01} ; y_1y_{10}, y_1y_{11} ; $y_{01}y_{010}, y_{01}y_{011}$; $y_{10}y_{100}, y_{10}y_{101}$; $y_{11}y_{110}, y_{11}y_{111}$
Z_2 (13 vertices, 12 edges)	the same with z in place of y
join edges	$x_{00}y_0$, $x_{00}z_0$

TABLE 2. The 24 arcs, with the authors' own symbol names and the printed Table-3 voltages. Reading the symbols in the order $a, b, c, \dots, \varepsilon$ reproduces both printed voltage rows.

sym	arc	volt.	sym	arc	volt.
a	$x_{0100} \rightarrow y_{010}$	+1	b	$x_{0100} \rightarrow z_{110}$	+2
c	$x_{0101} \rightarrow y_{011}$	+2	d	$x_{0101} \rightarrow z_{111}$	+1
e	$x_{0110} \rightarrow y_{100}$	+1	f	$x_{0110} \rightarrow z_{100}$	-1
g	$x_{0111} \rightarrow y_{101}$	+2	h	$x_{0111} \rightarrow z_{101}$	+3
j	$x_{1000} \rightarrow y_{010}$	+2	k	$x_{1000} \rightarrow z_{010}$	-1
l	$x_{1001} \rightarrow y_{011}$	+1	p	$x_{1001} \rightarrow z_{011}$	+3
q	$x_{1010} \rightarrow y_{110}$	+1	r	$x_{1010} \rightarrow z_{100}$	+1
s	$x_{1011} \rightarrow y_{111}$	+2	t	$x_{1011} \rightarrow z_{101}$	-1
u	$x_{1100} \rightarrow y_{100}$	+2	v	$x_{1100} \rightarrow z_{010}$	-3
w	$x_{1101} \rightarrow y_{101}$	+1	α	$x_{1101} \rightarrow z_{011}$	-2
β	$x_{1110} \rightarrow y_{110}$	+3	γ	$x_{1110} \rightarrow z_{110}$	-1
δ	$x_{1111} \rightarrow y_{111}$	-1	ε	$x_{1111} \rightarrow z_{111}$	+2

printed voltage rows exactly — $x \rightarrow y$: 1, 2, 1, 2, 2, 1, 1, 2, 2, 1, 3, -1 and $x \rightarrow z$: 2, 1, -1, 3, -1, 3, 1, -1, -3, -2, -1, 2 — which is an internal checksum on the table.

Remark (a forced correction to the published Table 3). The **eighth** “Starting leaf” entry is printed x_{1001} in both sub-rows, duplicating the sixth; it must read x_{1011} , as in Table 2. This is a defect of the published paper, present in the preprint .tex and on p. 26 of the version of record alike. Three independent confirmations: x_{1011} is the only one of the twelve 4-bit x -leaf labels otherwise missing; the paper’s own commented-out earlier version of the same assignment, in the preprint’s source, reads “ $(x_{1011}y_{111}) = 3$, $(x_{1011}z_{101}) = 9$ ”, and y_{111}, z_{101} are exactly the printed eighth “End” entries; and under the literal printed reading x_{1001} would have degree 5 and x_{1011} degree 1, so the object would not be the one Theorem 5.5 describes, whereas

with the correction every non-pinned vertex has degree 3 and every y - and z -leaf receives exactly two arcs. All numbers below use the corrected reading.

With Table 2 in place, G_{12} has $|V| = 52$, $|E| = 75$, cycle rank $\beta = 75 - 52 + 1 = 24$, degree multiset $\{1^3, 3^{49}\}$ with the three degree-1 vertices exactly x^*, y^*, z^* ; it is connected, bipartite, and has no 4-cycles. All three pinned-to-pinned tree distances equal 6, so every lift already contains a 12-cycle: “girth 12” below means exactly “no cycle shorter than 12”, never “girth at least 12”.

3. THE DECISION CRITERION

Let v assign an element of $\mathbb{Z}_m$ to each of the 24 arcs, tree edges being 0, and let $\mathcal{C}$ be the set of simple cycles of G_{12} of length at most 10. Traversing a cycle, an arc contributes $+v$ if crossed in its direction and $-v$ otherwise.

Lemma 3. *The lift $(G_{12}; m)_v$ has girth 12 if and only if $\sum_{e \in C} \pm v(e) \not\equiv 0 \pmod{m}$ for every $C \in \mathcal{C}$.*

Proof. Necessity is elementary: if a simple cycle C of length $L \leq 10$ has $\sum_C v \equiv 0$, then C lifts to a simple L -cycle, and $L < 12$.

For sufficiency, three families of base walks could lift to a short cycle: simple cycles, which the condition handles; lollipop walks through a pinned vertex; and non-reversing non-cycle walks. The proof of Theorem 5.5 in [1, p. 25] disposes of the last two, verbatim: “*By construction of T_{12} , all lollipop walks lift to long cycles. Next, we need to consider non-reversing non-cycle walks in the voltage graph that may lift to short cycles. From Observation [nonCycleWalks], any such walk must involve at least one 4-cycle. However, analysis of the cycles in G_{12} shows that, in fact, there are no 4-cycles (by construction), so there are no short walks of this sort that lift to short cycles.*” Both disposals are properties of the *arc set alone*, so they hold for every assignment on those arcs and not only for the printed one. Directly: a pinned-vertex lollipop with tree-stem length d and loop length c lifts to a cycle of length $2d + c + 2 \geq 12$, the two minima being $(d, c) = (1, 8)$ via the 8-cycle $y_0 y_{01} y_{010} x_{0100} x_{010} x_{01} x_0 x_{00} y_0$ and $(d, c) = (2, 6)$, each giving exactly 12; and G_{12} has no 4-cycle. $\square$

Only necessity is used for the nonexistence proofs, so exhausting the assignment space against the condition and finding nothing *proves* nonexistence: dropping a walk family can only weaken the constraint set, which is safe for a NO. Theorem 1 does not rely on Lemma 3 at all — every witness in Section 4 is verified by building the full lift and computing its girth.

Two facts about $\mathcal{C}$ and about the parametrisation:

Proposition 4. $|\mathcal{C}| = 126$, with 12 cycles of length 6, 24 of length 8 and 90 of length 10; counted with both orientations this is 252, the number published in the proof of Theorem 5.5 (“There are 252 short cycles that are not formed by going back and forth along a single edge”).

Proposition 5. *The 24 arc voltages are a complete parametrisation of the $\mathbb{Z}_m$ lifts of G_{12} up to isomorphism. Consequently the $m = 4$ and $m = 5$ cells are exactly $\mathbb{Z}_4^{24}$ and $\mathbb{Z}_5^{24}$.*

Proof. This is the standard spanning-tree normalisation for voltage graphs (Gross and Tucker), applied here. A pinned vertex is joined to all m copies of its neighbour, so its edge voltage is irrelevant: that fixes 3 of the 51 tree edges. The gauge move $v(e) \mapsto v(e) + f(\text{head}) - f(\text{tail})$, for $f : V_{\text{ord}} \rightarrow \mathbb{Z}_m$, induces the lift isomorphism $w^i \mapsto w^{i+f(w)}$; the 49 ordinary vertices span a tree with 48 edges, and choosing f along that tree sets all 48 of those voltages to 0. Since $48 + 3 = 51 = |E(T_{12})|$, all tree voltages may be taken 0 and the remaining $\beta = 24$ arc voltages are free. $\square$

Of this proposition the program of Section 7 checks only the counts 49, 48 and $48 + 3 = 51 = |E(T_{12})|$; the normalisation argument itself is the standard one just cited and is not re-verified there. Nothing in Theorem 2 depends on the “up to isomorphism” clause: the two nonexistence proofs quantify over *all* of $\mathbb{Z}_4^{24}$ and $\mathbb{Z}_5^{24}$ with no quotient taken.

Exactly 14 of the 24 arcs are forced nonzero, namely those whose fundamental cycle has length at most 10: $a, b, c, d, e, f, g, h, j, k, l, p, v, \alpha$, with a and c at length 8 and the other twelve at length 10. The remaining ten, $q, r, s, t, u, w, \beta, \gamma, \delta, \varepsilon$, have fundamental cycles of length 12 and may legitimately be 0; zero entries do occur in the witnesses below.

4. EXISTENCE AT $m = 6, 7, 8$

Each witness is 24 residues, listed in the symbol order of Table 2. The base graph is Section 2 and the lift is the standard $\mathbb{Z}_m$ derived graph, so a reader needs nothing further.

$m = 6$, a $(\{3, 6\}; 12)$ -graph of order 297.

$a=1, b=3, c=3, d=4, e=2, f=1, g=4, h=2, j=2, k=3, l=1, p=5,$
 $q=3, r=3, s=2, t=5, u=5, v=1, w=3, \alpha=2, \beta=0, \gamma=5, \delta=1, \varepsilon=3.$

$m = 7$, a $(\{3, 7\}; 12)$ -graph of order 346.

$a=1, b=3, c=3, d=4, e=4, f=1, g=6, h=2, j=2, k=3, l=1, p=4,$
 $q=2, r=0, s=4, t=4, u=2, v=1, w=0, \alpha=5, \beta=3, \gamma=0, \delta=6, \varepsilon=5.$

$m = 8$, a $(\{3, 8\}; 12)$ -graph of order 395: the $m = 7$ vector read in $\mathbb{Z}_8$, with $g = 7$ in place of 6.

In each case the lift has $49m + 3$ vertices and $75m$ edges, three vertices of degree m and $49m$ of degree 3, and girth exactly 12. The three occurrences of 0 at $m = 7, 8$ are legal: r, w, γ are among the ten arcs whose fundamental cycles have length 12.

5. NONEXISTENCE AT $m = 4$: A PIGEONHOLE, WITH NO COMPUTATION

Proposition 6. *No $\mathbb{Z}_4$ voltage assignment on the 24 arcs of G_{12} has a lift of girth 12. The same argument re-proves the case $m = 3$.*

Proof. Consider the four arcs into z_{010} and z_{011} , in the notation of Table 2:

$$\begin{aligned} k &= v(x_{1000} \rightarrow z_{010}), & v &= v(x_{1100} \rightarrow z_{010}), \\ p &= v(x_{1001} \rightarrow z_{011}), & \alpha &= v(x_{1101} \rightarrow z_{011}). \end{aligned}$$

(i) *Each is nonzero.* In T_{12} we have $d(x_{1000}, z_{010}) = 9$: six steps up to x_{00} , then three down, along

$$x_{1000} \ x_{100} \ x_{10} \ x_1 \ x \ x_0 \ x_{00} \ z_0 \ z_{01} \ z_{010}.$$

So the arc k closes a cycle of length $1 + 9 = 10 \leq 10$ whose only arc is k , namely

$$x \ x_0 \ x_{00} \ z_0 \ z_{01} \ z_{010} \ x_{1000} \ x_{100} \ x_{10} \ x_1 \ x.$$

By Lemma 3 (necessity), $k \neq 0$. The three other distances are 9 as well, so $p, v, \alpha \neq 0$.

(ii) *They are pairwise distinct.* Each of the six pairs is joined by a cycle of length at most 10 traversing one of the two arcs forwards and the other backwards, using $d(x_{1000}, x_{1001}) = d(x_{1100}, x_{1101}) = 2$, the four cross distances = 6, and $d(z_{010}, z_{011}) = 2$:

pair	length	cycle
k, p	6	$x_{100} \ x_{1000} \ z_{010} \ z_{01} \ z_{011} \ x_{1001}$
v, α	6	$x_{110} \ x_{1100} \ z_{010} \ z_{01} \ z_{011} \ x_{1101}$
k, v	8	$x_1 \ x_{10} \ x_{100} \ x_{1000} \ z_{010} \ x_{1100} \ x_{110} \ x_{11}$
p, α	8	$x_1 \ x_{10} \ x_{100} \ x_{1001} \ z_{011} \ x_{1101} \ x_{110} \ x_{11}$
k, α	10	$x_1 \ x_{10} \ x_{100} \ x_{1000} \ z_{010} \ z_{01} \ z_{011} \ x_{1101} \ x_{110} \ x_{11}$
p, v	10	$x_1 \ x_{10} \ x_{100} \ x_{1001} \ z_{011} \ z_{01} \ z_{010} \ x_{1100} \ x_{110} \ x_{11}$

Each such cycle has voltage sum $\pm(v_1 - v_2)$ for the two arcs concerned, so necessity gives $k \neq p$, $k \neq v$, $k \neq \alpha$, $p \neq v$, $p \neq \alpha$, $v \neq \alpha$ in $\mathbb{Z}_m$.

(iii) Hence k, p, v, α are four pairwise distinct nonzero elements of $\mathbb{Z}_m$, which requires $m - 1 \geq 4$. But $\mathbb{Z}_4$ has only three nonzero residues, and $\mathbb{Z}_3$ only two. Pigeonhole. $\square$

6. NONEXISTENCE AT $m = 5$

Proposition 7. *No $\mathbb{Z}_5$ voltage assignment on the 24 arcs of G_{12} has a lift of girth 12.*

The cell is $\mathbb{Z}_5^{24}$ by Proposition 5, that is $5^{24} = 59,604,644,775,390,625$ points. Branch-and-prune against the 126 necessary conditions of Lemma 3, over the *complete* space with no gauge quotient and no symmetry break of any kind, exhausts it and returns no solution. Only necessity is used, so the negative cannot be an artefact of an over-strong constraint set. This case is machine-only here: we know of no short hand argument for it.

7. THE ACCOMPANYING PROGRAM

This folder contains `verify.py` and `verify.output.txt`, a recorded run of it: Python 3.9, standard library only, no solver and no external data file, 80 checks, all passing. It is built from Section 2 alone and does not parse this note. The quantities transcribed out of the note and re-derived there are: the tree by two independent constructions, and the invariants of Section 2; the census of Proposition 4 by two algorithms (path-DFS with min-index canonicalisation, and XOR of fundamental cycles over the co-tree arcs), agreeing on edge sets; the impossibility of the literal printed Table 3 (Remark 2); published invariants of [1] — the “252 short cycles” as 126×2 , the absolute voltage sums lying in “ $\{1, 2, \dots, 8\}$ ”, the absence of 4-cycles, the order $49m + 3$ with three vertices of degree m , and the girth profile of the printed table reported in Section 1; the girth of each witness of Section 4 on the full lift, by two algorithms (BFS from every vertex, and deleting each edge and BFS-ing between its endpoints); the ten constraints of Proposition 6, each identified with one of the 126 enumerated cycle forms; and the exhaustions of $\mathbb{Z}_4^{24}$ and $\mathbb{Z}_5^{24}$, all 24 variables branched, all 126 forms enforced, no gauge quotient and no symmetry break.

What it does not do. It does not re-verify the normalisation argument of Proposition 5, only its counts; it says nothing about the wide variant of Section 8; and it makes no statement about published order records, of which this note claims none. The recorded run also prints checks on matters this note makes no claim about, and it makes no claim to have enumerated everything stated here.

8. WHAT IS NOT SETTLED

- (1) **The wide reading is open at $m = 4$ and $m = 5$, and which reading the sub-bullet intends is not settled here.** Whether a *different* pairing of the twelve x -leaves to the y - and z -leaves — one of each, no 4-cycles — admits a girth-12 $\mathbb{Z}_4$ or $\mathbb{Z}_5$ assignment is untouched. In favour of the narrow reading, the sibling sub-bullets of [1, p. 28–29] spell the wide variant out explicitly (“*either a different voltage assignment for the same arcs as in H_{10} , or a different assignment of arcs and voltages using the same ‘underlying’ tree*”) and ours contains no such device; but that is an inference about wording, not an author statement, so a reader who prefers the wide reading should treat Theorem 2 as a statement about the G_{12} arc set only, and the wide question at $m = 4, 5$ as open. The issue does not arise at $m = 6, 7, 8$, where Theorem 1 answers the sub-bullet on either reading.
- (2) **Nothing is claimed about $m \geq 9$** beyond reproducing Theorem 5.5, nor about the sibling H_{10} and H_{12} sub-bullets of the same open-questions section, nor about the $(3, 14)$ -cage bullet, nor about any order record or upper bound at any m .

- (3) **The Table-3 correction is ours.** Remark 2 is a defect of the published paper; the correction is forced three ways, but it is a correction and a referee should see it as such.

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